

Teach or Elicit? Why the RLVR Capability-Ceiling Debate Resolves Into a Demand for Proprietary Vertical Data

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Abstract

A live 2025 debate divides the reinforcement-learning-from-verifiable-rewards (RLVR) community. One camp [2504.13837] shows that RLVR mainly sharpens sampling efficiency over reasoning paths the base model can already reach: RL-trained models win at small-k pass@1 but base models overtake them at large pass@k, so RL narrows rather than expands the reasoning boundary. A second camp [2505.24864] shows that prolonged RL with KL control and reference-policy resets discovers strategies the base model never reaches even under exhaustive sampling, including tasks where the base fails entirely at any number of attempts while the RL model reaches a 100% pass rate. We argue this dispute, however resolved, points to the same economic conclusion. In the elicitation view, the capability ceiling is set by the base model's pretraining distribution; in the expansion view, the gains concentrate exactly where the base is weakest. Both make new, unseen data the binding constraint. Synthetic self-improvement loops cannot lift that ceiling because recursive training erases the distribution's rare tails [2305.17493]. The only durable source of genuinely new signal is human-grounded data from under-documented vertical workflows, where domain corpora demonstrably lift models to expert level [2305.09617]. We quantify the debate with verbatim benchmark numbers, formalize a teach-vs-elicite decomposition, and derive a vertical-data investment thesis: target verticals whose capability is absent from pretraining and whose outcomes are verifiable.

Introduction

Reinforcement learning with verifiable rewards (RLVR) is now the core recipe behind frontier reasoning models. DeepSeek-R1-Zero is trained via large-scale RL with no supervised fine-tuning yet develops emergent self-reflection and verification, and DeepSeek-R1 reaches 79.8% pass@1 on AIME 2024, 97.3% on MATH-500, and a 2029 Codeforces Elo (96.3rd percentile) [1]. Open RLVR systems match this frontier efficiently: DAPO reaches 50 points on AIME 2024 with a Qwen2.5-32B base, beating DeepSeek-R1-Zero-Qwen-32B (47 points) while using 50% of the training steps [2]. Since R1, the recipe has only scaled: by late 2025 the first systematic >400,000-GPU-hour study of RL fitted predictable sigmoidal compute-performance curves and a best-practice recipe (ScaleRL), bringing RL training close to the predictability long achieved in pre-training [17]. Scaling post-training RL compute now carries open models to the closed frontier — DeepSeek-V3.2 performs comparably to GPT-5, and its high-compute Speciale variant surpasses GPT-5, matches Gemini-3.0-Pro reasoning, and won gold at both IMO 2025 and IOI 2025 [18]. This trajectory makes the underlying question sharper, not moot: as RL compute scales predictably, what ultimately caps the capability it can buy?

Beneath these headline wins sits an unresolved scientific question with direct strategic consequences: does RLVR **teach** the model new reasoning capability, or merely **elicit** capability the base model already latently holds? In 2025 the field split. One camp [3] measured pass@k at large k and concluded RL narrows the reasoning boundary; another [4] showed prolonged RL expands it into genuinely novel strategies. This paper takes the position that the answer, in either direction, converts into the same data-strategy conclusion.

Our thesis: the capability ceiling of an RLVR-trained model is set by its base model's data, synthetic loops cannot raise that ceiling because they collapse the distribution's tails [5], and the only source of genuinely new signal is human-grounded data from proprietary vertical workflows [6]. The training-method debate, followed to its end, is an argument for vertical-data acquisition. We make this argument quantitatively, copying benchmark numbers verbatim from primary sources, and we explicitly flag the counter-evidence and the limits of the inference.

Background and Related Work

RLVR descends from RLHF [1707.06347, 2203.02155] but replaces a learned human-preference reward with a programmatic checker: a math answer that verifies, code that passes tests, an instruction precisely followed. Ouyang et al. established the SFT to reward-model to PPO pipeline; the verifiable-reward turn removed the human reward model for tasks with checkable answers and was named RLVR by the Tulu-3 work [7].

Process-level supervision sharpened the recipe: Lightman et al. found a process-supervised reward model solves 78% of a representative MATH subset and that process supervision significantly outperforms outcome supervision, with the gap widening as more solutions are sampled [8]. This established that *where* the verifiable signal is applied (every step vs. final answer) materially changes capability gains.

The teach-vs-elicitation question is the natural next probe. If RL only re-weights existing trajectories, the base model's support bounds achievable capability; if RL expands support, RL itself can be a capability source. We review the two primary 2025 results that anchor the debate in Section 4, and the model-collapse literature [5, 2404.01413] that closes the synthetic-data escape hatch in Section 6.

Methodology and Approach

This is a synthesis paper: our method is evidence triangulation across four primary sources plus corroborating benchmarks, not a new training run. We adopt three rules to keep the inference rigorous. First, every quantitative claim is copied verbatim from a primary source and tagged with its citation; we never interpolate empirical numbers. Second, we treat the teach-vs-elicitation dispute as a genuine open question and require our strategic conclusion to hold under *both* resolutions, not just the one favorable to the thesis. Third, we separate empirical findings (benchmark numbers), projections (data-exhaustion forecasts, labeled estimates), and constructed strategic figures (the investment matrix, labeled analysis).

We formalize the debate with a pass@k decomposition. Let $S_{\text{base}}(k)$ be the probability the base model produces a correct solution within k samples, and $S_{\text{RL}}(k)$ the same for the RL-trained policy. The *elicitation* hypothesis predicts $S_{\text{RL}}(1) > S_{\text{base}}(1)$ but a crossover k^* such that $S_{\text{base}}(k) \geq S_{\text{RL}}(k)$ for $k \geq k^*$ — RL concentrates mass on already-reachable solutions. The *expansion* hypothesis predicts the existence of problems where $S_{\text{base}}(k) = 0$ for all feasible k while $S_{\text{RL}}(k) > 0$ — RL reaches outside the base support. These are directly measurable and the two camps measured them; we report both in Section 4.

To test whether the resolution matters for data strategy, we trace each hypothesis to its implied binding constraint. Under elicitation, the ceiling is the base distribution's support, fixed by pretraining data. Under expansion, the gains concentrate where the base is weakest — again a statement about pretraining-data coverage. We then ask whether synthetic data can relax either constraint (Section 6) and where non-synthetic new signal is found (Section 7).

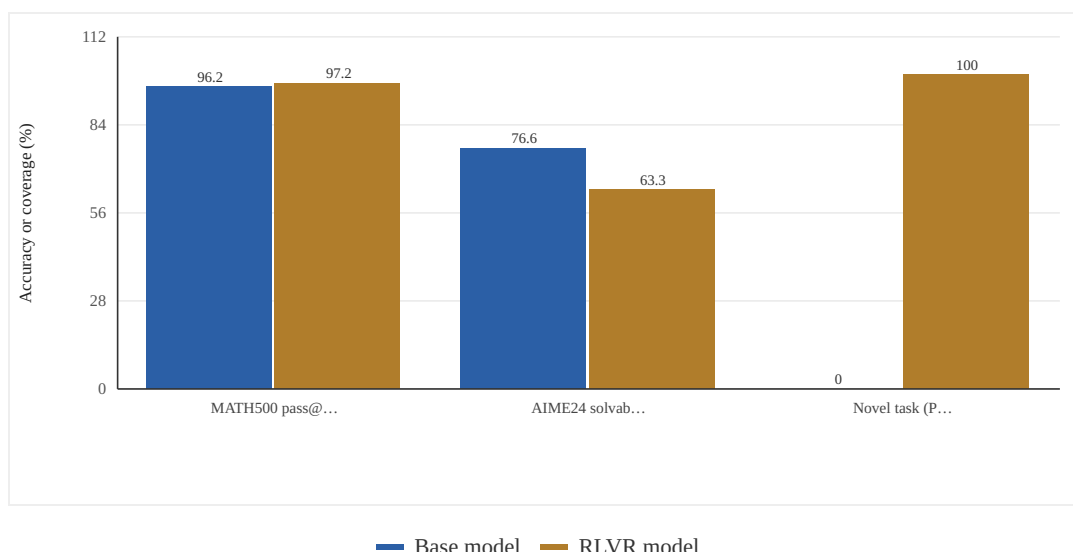


Figure 1. RLVR sharpens small-k accuracy but the large-k reasoning boundary is contested. On MATH500 the base nearly matches the RL model at pass@256 (96.2% vs 97.2%) [3]. On AIME24 at k=1024, 63.3% of problems are solvable by both models while the base additionally solves 13.3% that the RL model loses — so the base reaches 76.6% (63.3+13.3) of base-or-RL-solvable problems versus the RL model's 63.3%, the elicitation camp's narrowing evidence [3]. On ProRL's novel tasks the base fails entirely at any k while the RL model reaches a 100% pass rate — the expansion camp's evidence [4]. The same metric supports opposite readings depending on domain coverage. *Yue et al. 2025 (arXiv:2504.13837), HTML v3, Table 2 (AIME24 k=1024 Venn coverage; MATH500 pass@256); Liu et al. (NVIDIA ProRL) 2025 (arXiv:2505.24864).* · Empirical data.

Evidence: The Teach-vs-Elicit Split

The elicitation camp [3] (Yue et al., NeurIPS 2025 Oral; ICML 2025 AI4MATH Workshop Best Paper) measured pass@k across model families and benchmarks, using zero-RL models from SimpleRLZoo on Qwen2.5 bases. Their central finding is a crossover: RL-trained models win at small k (e.g., pass@1) but base models achieve a higher pass@k once k is large. On AIME24 at k=1024, a Venn decomposition shows 63.3% of problems are solvable by *both* the base and RL model while 13.3% are solvable by the *base alone* (and lost to the RL model) — direct evidence that RL prunes reachable solutions rather than adding them [3]. On MATH500 the base reaches 96.2% pass@256 versus the RL model's 97.2%, a narrow gap, with the base overtaking on harder benchmarks at larger k [3]. Their conclusion, verbatim: RLVR 'improves sampling efficiency toward already-reachable correct paths but does not elicit fundamentally new reasoning patterns,' and 'the reasoning capability boundary of LLMs often narrows as RLVR training progresses' [3]. Notably, they find distillation *can* introduce new patterns where RLVR cannot.

The expansion camp [4] (NVIDIA ProRL) used prolonged RL (>2,000 steps) with KL-divergence control and reference-policy resetting on a diverse task suite. Their Nemotron-Research-Reasoning-Qwen-1.5B shows average pass@1 improvements of 14.7% on math and 13.9% on coding over DeepSeek-R1-Distill-Qwen-1.5B, plus +54.8% on logic puzzles, +25.1% on STEM reasoning, and +18.1% on instruction-following [4]. Critically, they report tasks 'where base models fail entirely regardless of the number of attempts' (pass@k = 0 at any k) while the RL model reaches a 100% pass rate — direct evidence of support expansion that the elicitation framing cannot accommodate [4].

The two results are not contradictory once we read them as bounding the same quantity from opposite sides. Yue et al. study standard-duration RL on math benchmarks the base already covers; ProRL studies prolonged RL on tasks where reasoning-boundary improvements correlate with the base model's weakness and with training duration

— i.e. gains are largest exactly where base competence is *lowest* [4]. The disagreement is about *how far* RL can push, not about *what governs* the push: base-model coverage.

Table 1. Both camps' headline numbers, copied verbatim. The split is about how far RL can push (narrowing vs expansion), not about what governs the push (base-model coverage). Note: the AIME24 63.3/13.3 figures are a k=1024 coverage decomposition, not single-sample accuracies.

Paper	Claim	Key metric	Number
Yue et al. [3] (2504.13837)	RL prunes base-reachable solutions	AIME24 coverage, Qwen2.5 base vs RL (k=1024)	63.3% solved by both; 13.3% by base alone (lost by RL)
Yue et al. [3]	Base nearly matches RL at high k	MATH500 pass@256, base vs RL	96.2% vs 97.2%
ProRL [4] (2505.24864)	RL expands boundary	Avg pass@1 gain, math / code	+14.7% / +13.9%
ProRL [4]	Gains largest where base weak	Logic puzzles / STEM / instruction-following	+54.8% / +25.1% / +18.1%
ProRL [4]	RL reaches outside base support	Novel tasks where base fails entirely	base fails at any k → RL 100% pass rate
Spurious Rewards [9] (2506.10947)	RLVR elicits latent skill	Qwen2.5-Math-7B MATH-500 gain: random / incorrect / GT	+21.4% / +24.1% / +29.1%
Spurious Rewards [9]	Gain is data/model-dependent	'Code reasoning' frequency after RLVR	65% → >90% (fails on Llama3/OLMo2)

arXiv:2504.13837 (HTML v3, Table 2); arXiv:2505.24864 + NVIDIA model card; arXiv:2506.10947.

Table 2. Synthetic loops collapse the rare tails that hold under-documented vertical capability; the durable anchor is human-grounded vertical data, which demonstrably lifts models to expert level.

Source	Mechanism / result	Quantified finding
Shumailov et al., Nature 2024 [5] (2305.17493)	Recursive synthetic training erases tails	OPT-125M / wikitext: tails disappear, output → shrunken low-variance distribution, perplexity rises across generations
Gerstgrasser et al. [2404.01413]	Accumulating real + synthetic avoids collapse	Real-data anchor required to prevent tail loss
Singh et al. ReST^EM [2312.06585]	Self-training needs external correctness filter	Surpasses human-data finetuning only on verifiable tasks
Med-PaLM 2 [6] (2305.09617)	Domain corpus → expert level	MedQA 86.5% (+19% over Med-PaLM), new SOTA
MEDITRON-70B [10] (2311.16079)	Curated corpus is the lever	Within 5% of GPT-4, 10% of Med-PaLM-2 on medical benchmarks

Nature 2024 (s41586-024-07566-y); arXiv:2404.01413, 2312.06585, 2305.09617, 2311.16079.

Table 3. Public human text is finite and nearly exhausted; verifiable environments manufacture fresh gradeable signal on demand, making them a durable data asset rather than a finite stock.

Source	Finding	Number
Villalobos et al. / Epoch AI [11] (2211.04325)	Effective public text stock	~300T tokens (90% CI 100T–1000T)
Villalobos et al. [11]	Exhaustion window	Public human text fully utilized 2026–2032
Hernandez et al. [2205.10487]	Repetition is not a workaround	Beyond ~4 epochs marginal value decays; damages induction heads
DeepSeek-R1 [1] (2501.12948)	RLVR frontier on verifiable tasks	AIME24 79.8%, MATH-500 97.3%, Codeforces 2029 Elo (96.3 pct)
DAPO [2] (2503.14476)	Open RLVR matches frontier efficiently	AIME24 50 pts vs R1-Zero-Qwen-32B 47, at 50% of training steps

arXiv:2211.04325 / Epoch AI; arXiv:2205.10487, 2501.12948, 2503.14476.

Analysis: A Shared Dependence on Base-Model Data

Both camps locate the lever in the base model's data distribution. Under elicitation, capability is capped by the base support, which is fixed by pretraining tokens. Under expansion, the marginal capability RL adds is largest where the base is weakest — ProRL's own finding that boundary expansion is greatest on the domains where the base is least competent [4] says the headroom RL can exploit is a function of where pretraining coverage is thin. Either way, the binding constraint on **new** capability is **data the base model has not absorbed**.

A third, independent result tightens this. The spurious-rewards study [9] found that on Qwen2.5-Math-7B, RLVR improves MATH-500 by 21.4% with a **random** reward, 13.8% with a format reward, 24.1% with an **incorrect-label** reward, and 27.1% with majority voting — nearly matching the 29.1% from ground-truth rewards [9]. The mechanism is elicitation of a latent, pretraining-acquired behavior: 'code reasoning' frequency rises from 65% to over 90% after RLVR even with spurious rewards [9]. Decisively, 'spurious rewards that work for Qwen often fail to yield gains' on Llama3 or OLMo2 [9] — the gain is model- (hence data-) dependent, not reward-dependent. Where the latent skill is absent from pretraining, no clever reward conjures it.

This yields the operative inference: RLVR's value as a **capability creator** (not merely a sharpener) is concentrated in domains the base model never saw. In math and code — heavily represented in pretraining — much of the gain is elicitation of existing skill (the spurious-reward result is the extreme demonstration). In under-documented verticals, the base support is genuinely thin, so the marginal value of new verifiable data is highest there.

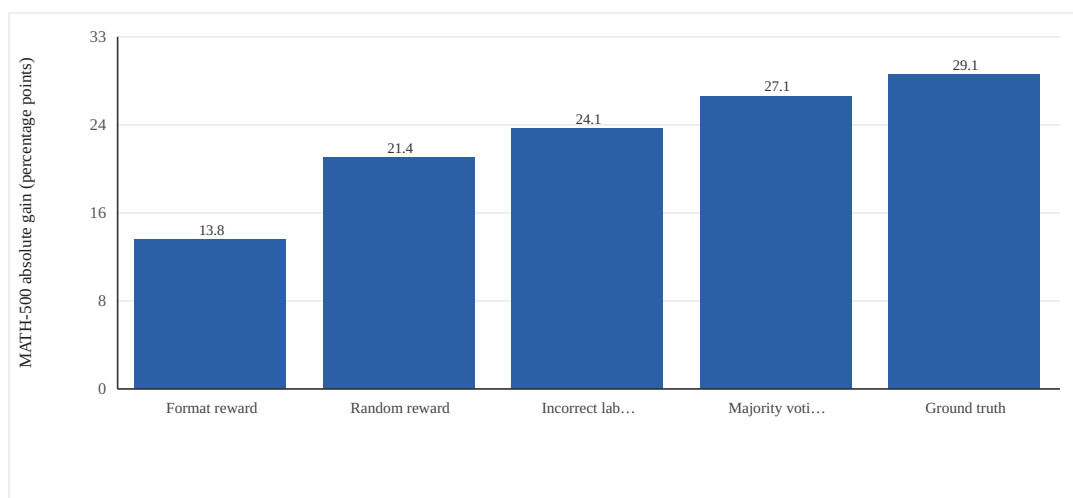


Figure 2. On Qwen2.5-Math, even spurious rewards nearly match ground truth — RLVR elicits latent skill. RLVR improves Qwen2.5-Math-7B on MATH-500 by 21.4% with a random reward and 24.1% with an incorrect-label reward, nearly matching the 29.1% from ground-truth rewards. The gain is elicitation of a latent (pretraining-acquired) behavior, not teaching — 'code reasoning' frequency rises from 65% to over 90% — and it fails to transfer to Llama3/OLMo2 [9]. Where the skill is absent from pretraining, no reward conjures it. *Shao, Li, Xin et al., 'Spurious Rewards: Rethinking Training Signals in RLVR' 2025 (arXiv:2506.10947). · Empirical data.*

Why Synthetic Data Cannot Lift the Ceiling

If new data is the constraint, the obvious escape is to manufacture it synthetically. The model-collapse literature closes this door for the case that matters. Shumailov et al. (Nature 2024) show that training on recursively generated data causes irreversible 'model collapse': 'the tails of the original content distribution disappear,' losing rare and unique elements, with successive generations converging toward a shrunken, low-variance output and rising perplexity in their OPT-125M / wikitext experiments [5].

This is precisely fatal to the vertical use-case, because the capability we need lives in the tails. The skills absent from pretraining are by definition rare relative to the open web; a synthetic loop trained on a model that lacks them cannot generate them and actively erodes whatever sliver it had. The known fix — accumulating synthetic data **alongside** real data rather than replacing it [2404.01413] — only works because a real-data anchor is retained. The anchor is the scarce, non-synthesizable input.

Self-improvement methods confirm the ceiling from the algorithmic side: ReST[^]EM works only because it filters self-generated samples by an **external** binary correctness signal [2312.06585], and self-rewarding is bounded by the model judging itself [2401.10020]. Every durable bootstrapping gain requires an external anchor — verifiable reward or human-grounded data. That anchor is what retains value in a synthetic-data-flooded economy.

Where the New Signal Lives: Vertical Workflows

The data the base models never saw is disproportionately in proprietary vertical workflows, and the vertical literature shows such data lifts models to expert level. Med-PaLM 2, via medical-domain fine-tuning, scored up to 86.5% on MedQA (USMLE-style), improving over Med-PaLM by over 19% and setting a new state-of-the-art [6]. MEDITRON-70B, through continued pretraining on a curated medical corpus, came within 5% of GPT-4 and 10% of Med-PaLM-2 on major medical benchmarks [10] — evidence that the curated corpus, not a novel method, is the lever.

Critically for the RLVR thesis, the highest-value verticals are those that *also* throw off verifiable outcomes: clinical claims/coding adjudication (codes that pass or fail an X12/CDT/ICD checker), insurance adjudication, lab/diagnostic workflows, contract review. These satisfy both conditions our analysis demands: capability absent from pretraining (so RL teaches rather than elicits) and a programmatic reward (so RLVR applies at all). Math and code satisfy verifiability but their data is public, so the base already covers them — exactly the regime where the spurious-reward result says gains are elicitation [9]. The frontier already instantiates this in the one vertical with abundant public verifiable traces — software: SWE-RL applies rule-based RL directly to raw GitHub software-evolution data (issues, commits, pull requests) and lifts a 70B model to a 41.0% solve rate on SWE-bench Verified, comparable to GPT-4o, while production agentic models such as Kimi K2 reach 65.8 on SWE-bench Verified by combining verifiable rewards with a self-critique rubric over real and synthetic environment interaction [19, 21]. These are existence proofs that RL over genuine workflow data trains strong agents; the open question for under-documented verticals is supplying the same verifiable workflow data where it is not already public.

This is the convergence of three literatures the individual papers never join: the training-method debate (teach vs elicit) points to base-model data; the data limit (collapse erases tails) forbids the synthetic shortcut; and the vertical wins show where non-synthetic new signal is found. The same bet — verifiable-reward environments in real verticals — is a training-method bet, a data-asset bet, and a vertical bet simultaneously.

The Macro Constraint: The Data Wall

The vertical-data thesis is reinforced by the exhaustion of public human text. Epoch AI estimates the effective stock of quality- and repetition-adjusted human-generated public text at roughly 300 trillion tokens (90% CI 100T–1000T) and forecasts that 'if current trends continue, LMs will fully utilize the stock of human-generated public text between 2026 and 2032 (or slightly earlier if models are overtrained)' [11]. Repeating data is not a workaround: beyond ~4 epochs the marginal value decays, and repetition damages copying and induction heads, shifting models toward memorization [2205.10487].

With the public stock running down and synthetic loops collapsing the tails, the only source of *fresh gradient signal* is interaction with verifiable environments — which manufacture gradeable signal on demand rather than drawing down a finite stock. This reframes 'environments with checkable outcomes' from a training trick into a durable, manufacturable data asset, and the data wall is the economic reason the timing is now. Recent results make this an empirical axis rather than a metaphor: scaling the *number* of verifiable environments (RLVE), not just compute, drives generalizable reasoning gains — joint training across 400 adaptive verifiable environments yields +3.37% absolute across six reasoning benchmarks from a strong 1.5B model, versus only +0.49% from continuing the original RL run with over 3x more compute [20]. Environments behave like a self-curating, manufacturable data source whose difficulty adapts to the policy, exactly the property a finite text stock lacks.

The teach-vs-elicitation debate and the data wall therefore reinforce each other. Even in the most conservative (pure-elicitation) reading of RLVR, raising the *base-model* ceiling requires new pretraining-grade data — and that data is exactly what is running out on the public web and disproportionately available only in proprietary verticals.

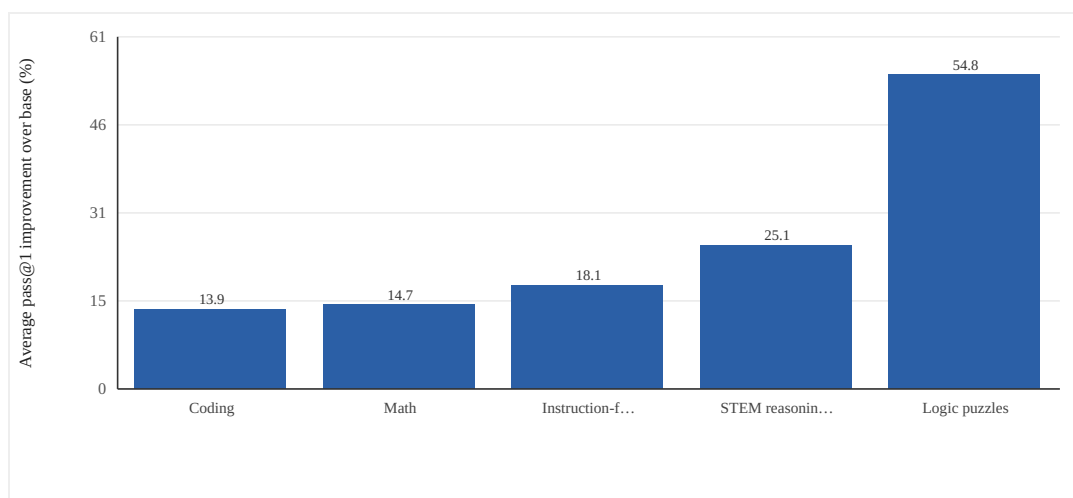


Figure 3. ProRL gains concentrate where base-model competence is lowest. ProRL's largest gains appear on logic puzzles (+54.8%) and STEM reasoning (+25.1%) — domains thin in the base distribution — and smallest on coding (+13.9%) and math (+14.7%), which are heavily pretrained. This is direct evidence that RL's capability-creation value concentrates where pretraining coverage is weakest, motivating investment in under-documented verticals. *Liu et al. (NVIDIA), 'ProRL' 2025 (arXiv:2505.24864); pass@1 gains per the official Nemotron-Research-Reasoning-Qwen-1.5B model card. · Empirical data.*

Implications for Data Strategy

The decomposition yields a concrete selection rule. Score a candidate vertical on two axes our evidence shows are jointly necessary: (1) base-model coverage gap — is the capability absent from pretraining, so created data **teaches**? and (2) outcome verifiability — does the workflow throw off a programmatic pass/fail, so RLVR applies? Verticals high on both (clinical claims/coding, insurance adjudication, lab/diagnostic) are where verifiable-reward data both teaches and is buildable. Verticals high only on verifiability (math, code) yield elicitation gains a competitor can also extract from public data — thinner moat.

Operationally, this argues for owning the real-data anchor and treating the RL method as a commodity. The spurious-reward and ProRL results together imply the **algorithm** is increasingly fungible (Yue et al. find a range of RLVR algorithms produce similar pass@k behavior [3]; even random rewards work on the right base [9]); the **signal source** is not. That RL compute now scales along predictable curves [17] and that gains track the **number** of verifiable environments [20] both push value away from the method and toward whoever owns the gradeable environments. Capital should concentrate on exclusive access to proprietary vertical corpora and the verifiers that grade them.

The recurring nature compounds the asset. Because the optimal data is the data the base never saw, and base models absorb more public data each generation, the coverage gap migrates — yesterday's proprietary edge can become tomorrow's pretrained commodity. This makes vertical data a renewing subscription surface, not a one-time dump: the buyer must keep re-acquiring signal at the receding frontier of base-model coverage.

Investment Thesis

For an investor, the through-line is that the AI value chain's binding constraint has migrated from compute to base-model coverage of new domains, and the only collapse-proof supply of that coverage is human-grounded, verifiable vertical data. Whoever owns exclusive access to such corpora — and the programmatic verifiers that turn them into reward signal — holds an asset that resists both method commoditization and synthetic substitution.

The asymmetry is favorable. The downside case (pure elicitation [3]) still requires new base-grade data to raise ceilings, so vertical data retains value. The upside case (expansion [4]) makes RL itself a capability multiplier on exactly the thin-coverage verticals where proprietary data lives, raising willingness-to-pay. There is no resolution of the debate under which proprietary verifiable vertical data loses value — a rare structural hedge.

The defensible play is therefore not a curation tool (commoditizing) nor a frontier-model bet (capital-intensive, winner-take-few) but ownership of the scarce input: real, verifiable, proprietary vertical data, priced as a renewing stream against the receding base-coverage frontier. Perpetual Labs' clinical-claims/coding bet sits squarely in the high-on-both-axes quadrant.

Limitations and Counter-Evidence

The teach-vs-elicitation debate is genuinely unresolved; we lean on it being unresolved rather than picking a winner, but if the elicitation camp [3] is fully correct, RLVR adds **no** new capability and the entire RL-as-teacher premise weakens — though our thesis survives because it then rests on the base-data ceiling, which still demands new vertical data. Conversely, if expansion [4] generalizes broadly, RL could extract more from existing data than we assume, modestly reducing the marginal value of new data.

Our benchmark numbers are model- and dataset-specific. The spurious-reward effect [9] is demonstrated primarily on Qwen2.5-Math and explicitly fails to transfer to Llama3/OLMo2 — it is evidence of model-dependence, not a universal law; subsequent work has also attributed part of the Qwen-specific effect to data contamination. The pass@k crossover [3] is sensitive to k, sampling temperature, and benchmark difficulty, and its headline AIME24 numbers (63.3% solved by both vs 13.3% by base alone) are a coverage decomposition at k=1024, not single-sample accuracies; the MATH500 gap (96.2% vs 97.2%) is narrow. These are real findings but should not be over-generalized.

The vertical evidence (Med-PaLM 2, MEDITRON) concerns medical QA, not the specific claims/coding RLVR environments we advocate; transfer is plausible but unproven. The data-exhaustion forecast [11] is a projection with a wide 90% CI (100T–1000T tokens) and assumes continued trend extrapolation. Finally, this analysis scores only the data-shape (coverage gap x verifiability); it does not weigh acquisition cost, regulatory constraints (e.g. HIPAA/data-resale rights), supply scale, or the behavior-cloning-vs-true-RL ceiling that a single-vertical bet faces. The go/no-go for any specific vertical requires that full diligence.

Conclusion

A narrow methodological dispute about whether RL teaches or elicits resolves, in either direction, into a statement about data: the capability ceiling traces to the base model's pretraining distribution, synthetic loops cannot raise it without collapsing the tails, and the only durable source of new signal is human-grounded verifiable data from under-documented verticals.

The numbers tell a consistent story. RL lifts small-k pass@1 yet can leave the reachable-solution set unchanged or smaller at large k (on AIME24, 13.3% of base-solvable problems become unsolvable after RL, while 63.3% remain solvable by both [3]); it can also reach problems the base never solves at any k while attaining a 100% pass rate [4]; and it can be driven by random rewards where the skill is already latent [9] but not where it is absent. All three observations point at base-model coverage as the lever, and at proprietary verticals as where that coverage is thinnest and most valuable to fill.

For a data-asset builder, the conclusion is sharp: buy the receding frontier of base-model coverage in verifiable verticals. That is the one asset whose value is invariant to how the teach-vs-elicited debate finally resolves.

INVESTMENT THESIS

The 2025 'does RL teach or just elicit?' debate has no resolution under which proprietary, verifiable vertical data loses value — a rare structural hedge. If RL only elicits (Yue et al.: RL wins small-k pass@1 but the base overtakes at large k — on AIME24 at k=1024, 63.3% of problems are solvable by both while RL loses 13.3% the base alone solves; random rewards nearly match ground truth on Qwen2.5-Math, +21.4% vs +29.1%), the capability ceiling is the base model's data and you must buy new base-grade signal. If RL expands (ProRL: base fails at any k while the RL model hits a 100% pass rate on novel tasks, gains largest on thin-coverage domains like logic +54.8%), the upside concentrates exactly on under-documented verticals. Synthetic loops can't substitute — recursive training collapses the rare tails (Nature 2024). As of late 2025 the recipe has only scaled — RL compute now follows predictable curves (ScaleRL, >400K GPU-hours [17]) and scaled-RL open models reach GPT-5/Gemini-3-Pro parity and IMO/IOI gold (DeepSeek-V3.2 [18]), while gains increasingly track the *number* of verifiable environments, not raw compute (RLVE: +3.37% vs +0.49% [20]) — which only sharpens the demand for owned, gradeable vertical environments. With public text exhausting 2026–2032 (~300T tokens), the defensible asset is exclusive access to real, verifiable vertical corpora plus their checkers, priced as a renewing stream against the receding frontier of base-model coverage. Concentrate capital on owning the signal source (clinical claims/coding, insurance, lab/diagnostic — high on both coverage-gap and verifiability), not the RL method, which the spurious-reward result shows is increasingly fungible.

References

- [1] DeepSeek-AI. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv:2501.12948, 2025. AIME 2024 pass@1 79.8%, MATH-500 97.3%, Codeforces 2029 Elo / 96.3rd percentile.
- [2] Yu, Q. et al. DAPO: An Open-Source LLM Reinforcement Learning System at Scale. arXiv:2503.14476, 2025. 50 points on AIME 2024 with Qwen2.5-32B base vs 47 for DeepSeek-R1-Zero-Qwen-32B at 50% of training steps.
- [3] Yue, Y. et al. Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? arXiv:2504.13837, 2025 (NeurIPS 2025 Oral; ICML 2025 AI4MATH Workshop Best Paper). Pass@k crossover; AIME24 k=1024 coverage 63.3% both / 13.3% base-only; MATH500 base 96.2% vs RL 97.2% pass@256.
- [4] Liu, M. et al. (NVIDIA). ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models. arXiv:2505.24864, 2025. Avg pass@1 gains +14.7% math, +13.9% code, +54.8% logic, +25.1% STEM, +18.1% instruction-following (Nemotron-Research-Reasoning-Qwen-1.5B model card); base fails at any k → RL 100% pass rate on novel tasks.
- [5] Shumailov, I. et al. AI models collapse when trained on recursively generated data. Nature 631, 755–759, 2024 (DOI 10.1038/s41586-024-07566-y; arXiv:2305.17493). Tails of the distribution disappear; OPT-125M / wikitext perplexity rises across recursive generations.
- [6] Singhal, K. et al. Towards Expert-Level Medical Question Answering with Large Language Models (Med-PaLM 2). arXiv:2305.09617, 2023 (Nature Medicine 2025). MedQA 86.5%, +19% over Med-PaLM.
- [7] Lambert, N. et al. Tulu 3 / RLVR: Reinforcement Learning with Verifiable Rewards. arXiv:2411.15124, 2024. Names and formalizes RLVR on tasks with checkable answers.
- [8] Lightman, H. et al. Let's Verify Step by Step. arXiv:2305.20050, 2023. Process-supervised reward model solves 78% of a MATH subset; process > outcome supervision, gap widens with more sampled solutions.
- [9] Shao, R., Li, S.S., Xin, R. et al. Spurious Rewards: Rethinking Training Signals in RLVR. arXiv:2506.10947, 2025. Qwen2.5-Math-7B MATH-500 gains: +21.4% random, +13.8% format, +24.1% incorrect, +27.1% majority, +29.1% ground truth; code-reasoning 65%→>90%; fails on Llama3/OLMo2.
- [10] Chen, Z. et al. MEDITRON-70B: Scaling Medical Pretraining for Large Language Models. arXiv:2311.16079, 2023. Within 5% of GPT-4 and 10% of Med-PaLM-2 on major medical benchmarks via curated-corpus continued pretraining.

- [11] Villalobos, P. et al. Will we run out of data? Limits of LLM scaling based on human-generated data. arXiv:2211.04325, 2022 / Epoch AI (ICML 2024). Effective stock ~300T tokens (90% CI 100T–1000T); exhausted 2026–2032.
- [12] Hernandez, D. et al. Scaling Laws and Interpretability of Learning from Repeated Data. arXiv:2205.10487, 2022. Double-descent from repetition; beyond ~4 epochs marginal value decays; damages copying/induction heads.
- [13] Gerstgrasser, M. et al. Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data. arXiv:2404.01413, 2024. Accumulating synthetic alongside real data avoids collapse.
- [14] Singh, A. et al. Beyond Human Data: Scaling Self-Training for Problem-Solving (ReST^{EM}). arXiv:2312.06585, 2023. Self-training works via external binary correctness filter; surpasses human-data finetuning on verifiable tasks.
- [15] Schulman, J. et al. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017. The default RL optimizer underlying RLHF/RLVR.
- [16] Ouyang, L. et al. Training language models to follow instructions with human feedback (InstructGPT). NeurIPS 2022 / arXiv:2203.02155. Canonical SFT→reward-model→PPO pipeline.
- [17] Khatri, D., Madaan, L. et al. (incl. R. Agarwal). The Art of Scaling Reinforcement Learning Compute for LLMs. arXiv:2510.13786, 2025. First >400,000-GPU-hour study fitting sigmoidal compute-performance curves for RL; proposes the ScaleRL best-practice recipe validated on a single run scaled to 100,000 GPU-hours. <https://arxiv.org/abs/2510.13786>
- [18] DeepSeek-AI. DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models. arXiv:2512.02556, 2025. Scaled RL post-training reaches parity with GPT-5; the high-compute DeepSeek-V3.2-Special surpases GPT-5, matches Gemini-3.0-Pro reasoning, and wins gold at IMO 2025 and IOI 2025. <https://arxiv.org/abs/2512.02556>
- [19] Kimi Team (Moonshot AI). Kimi K2: Open Agentic Intelligence. arXiv:2507.20534, 2025. 1T-total/32B-active MoE; joint RL with verifiable rewards plus a self-critique rubric over real and synthetic environment interaction; 65.8 SWE-bench Verified, 66.1 Tau2-Bench, 49.5 AIME 2025. <https://arxiv.org/abs/2507.20534>
- [20] Zeng, Z., Ivison, H., Wang, Y., Yuan, L. et al. RLVE: Scaling Up Reinforcement Learning for Language Models with Adaptive Verifiable Environments. arXiv:2511.07317, 2025. RLVE-Gym (400 procedurally-generated, difficulty-adaptive verifiable environments); environment scaling yields +3.37% absolute across six reasoning benchmarks vs +0.49% from continuing the original RL run with 3x more compute. <https://arxiv.org/abs/2511.07317>
- [21] Wei, Y., Duchenne, O., Copet, J., Synnaeve, G. et al. (Meta). SWE-RL: Advancing LLM Reasoning via Reinforcement Learning on Open Software Evolution. arXiv:2502.18449, 2025 (NeurIPS 2025). Rule-based RL over raw GitHub software-evolution data; Llama3-SWE-RL-70B reaches 41.0% on SWE-bench Verified, comparable to GPT-4o, with generalized out-of-domain reasoning gains. <https://arxiv.org/abs/2502.18449>